Item 1 Priority 1, Phase A

Reproduce BioLangFusion exactly: its encoders (NT v2 100M, RNA-FM, ESM-2 8M), codon grid, three heads, TextCNN; published splits, recipe. Fungal, E. coli, mRNA stability, Ab1, CoV-Vac. Gate: match the published scores or record each deviation. Tier 2, Gantt 17.

Item 1 Priority 1, Phase B

Change the encoders only: same splits, grid, heads, TextCNN and evaluation; one encoder swapped at a time (CaLM or other codon models, mRNA-FM, ProtBERT, other DNA encoders), then triples, on Fungal, E. coli, mRNA stability, Ab1 (CoV-Vac if feasible). Gantt 24.

Item 2 Priority 2

LucaOne, jointly pretrained and frozen; same folds and head on stability and GTEx. Verify the 1,280-token limit and sequence types first. Compare with the best single encoder and the best fusion; report cost and truncation. Tier 2, Gantt 41.

Item 3 Priority 3

Full-transcript half-life (Saluki human, mouse; LBKWK half-life). Gate: UTRs, CDS coordinates, labels, grouped splits. CDS only vs CDS + 3' UTR vs full transcript, + protein for the CDS; length and composition baselines. Tier 4, Gantt 42, conditional.

Item 4 Priority 4

Translation efficiency (TE human, mouse; MRL Sugimoto; LBKWK MRL), each target on its own split. 5' UTR, CDS, full transcript, + protein, three-encoder fusion; RiboNN as external benchmark; gain over best full transcript. Tier 4, Gantt 42, conditional.

Item 5 Low-cost control, in parallel: mRFP

Cached embeddings and labels in hand. Protein is identical by construction, so protein encoders are an invariant negative control; nucleotide encoders against GC3, codon-use and structure proxies. Tests representation specificity, not fusion. Tier 1, Gantt 18.

Item 6 Optional extension: structure

After the core benchmarks, only on a task where structure plausibly acts. Protein: sequence vs + ESMFold vs + AlphaFold2 features. RNA: sequence vs + predicted pairing (RNAfold vs EternaFold, MXfold2). Gains over sequence-only and simple structure features. Gantt 38, conditional.

Item 7 Exploratory: multi-teacher distillation

Six DNA, five RNA and four protein teachers (the twelve Stage 1 encoders plus two DNA and one RNA) distilled into one compact student per modality by matching each teacher's cosine-similarity matrix through its own projection, then the students fused. Think it through now; Gantt 43, conditional.

The research plan's seven items in its own order, on top of the Stage 3 and 4 design (Gantt chart as of October 15; dates subject to change). Coupled Mamba stays the priority architecture (Gantt 20 to 21). Every result reports Δfusion = Perf(fused) minus the best single encoder, confidence intervals, and the baselines B0 to B5.